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Enhanced oil recovery through wettability alteration using a novel TiO₂-SiO₂-tween 20 nano-fluid

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Outline

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Highlights

- Excellent enhancement of oil recovery through silica and titanium nano-particles with surfactant (Tween20).
- Employing low salinity (Losal) water solution in the nano-fluid for improving wettability alteration and IFT.
- Attaining an oil recovery of 29% in nano-fluid flooding process.
- In micromodel experiment, nano-fluid increased oil recovery by 34%.
- Flooding and micromodel approaches presented more realistic engineering evaluations.

Abstract

Chemical Enhanced Oil Recovery (CEOR) flooding is considered as one of the most effective techniques for extracting trapped oil from porous media. The use of nano-particles in EOR processes significantly aids in reducing interfacial tension (IFT) and modifying the wettability of reservoir rocks. In this regard, a blend of various nano-composites (NCs) yields better results than a single material, improving surface or interfacial properties. In this research, a new NC with low salinity (LoSal) water/engineered water was utilized as the base fluid to create smart one. Synthetic NC consists of titanium dioxide (TiO₂), silica (SiO₂), and Tween 20. In this regard, the synthesized NC was examined using various structural analyses, including EDX, XRD, FT-IR, and FESEM. To assess the effectiveness of the smart nano-fluid, a series of tests were conducted, including Zeta (ζ) potential, electrical conductivity, contact angle measurements, IFT assessments, core flooding experiments, and micromodel tests. Findings indicated that the smart nano-fluid mixture exhibited a superior synergistic effect, achieving a minimum IFT of 3.03 mN/m at a concentration of 500ppm. The contact angle measurements revealed a shift in rock wettability from oil-wet (OW) to water-wet (WW), decreasing from 159° to 30°. Additionally, the results showed that core flooding with the nano-fluid improves oil recovery by 29%. Also, the micromodel tests demonstrated a maximum oil recovery increase of 34% with the nano-fluid compared to LoSal water.

Keywords

Enhanced oil recovery; Nano-composites; Interfacial tension; Smart nano-fluid; Wettability alteration

5. Conclusions

CRedit authorship contribution statement

Declaration of competing interest

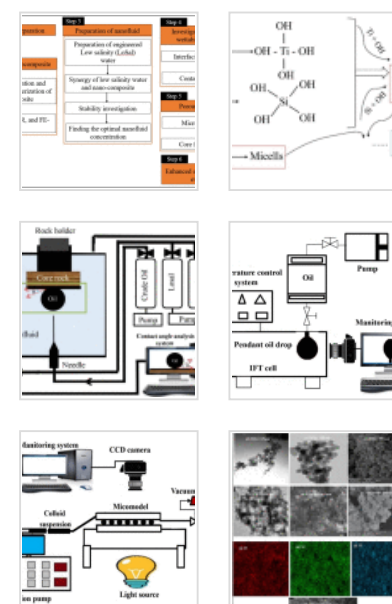
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Data availability

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Table 1. Key findings (IFT, contact angle, ...)

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Nomenclature

Table 2. Comparing the results of this pa...

CEOR

Chemical Enhanced Oil Recovery

EDAX

Energy dispersive X-ray

EOR

Enhanced Oil Recovery

FESEM

Field emission scanning electron microscopy

FT-IR

Fourier transform infrared

IFT

Interfacial tension

LoSal

Low salinity

NC

Nano-composite

NCs

Nano-composites

OW

Oil-wet

TEM

Transmission electron microscopy

WW

Water-wet

XRD

X-ray diffraction

 ζ

Zeta

1. Introduction

Oil and gas stored in underground porous media, characterized by extensive pores and interconnected throats, can be extracted using various enhanced oil recovery (EOR) methods such as water flooding, nano-particles, polymers, and surfactant flooding [[1], [2], [3]]. Traditional flow characteristic theories often stem from core flooding experiments; however, these experiments typically do not provide a clear understanding of the fluid transport processes within porous media [4,5]. Therefore, it is crucial to investigate the pore-scale physical processes of hydrocarbon extraction through experimental approaches to develop more efficient EOR techniques. By enhancing the macroscopic sweep efficiency of different crude oil types, chemical enhanced oil recovery (CEOR) significantly boosts oil production [6,7]. The CEOR works by changing the wettability of the porous media from oil-wet (OW) to water-wet (WW) [[8], [9], [10], [11]]. Among the various chemicals employed for this purpose, polymers and surfactants play an important role [12,13]. Recently, the use of surfactants as EOR agents has attracted the attention of researchers due to their substantial effect on increasing oil yields [14,15]. In addition, nano-particles have a significant role because of their unique properties and abilities to enhance CEOR performance [16,17]. These nano-particles exhibit distinctive physical and chemical characteristics though their high surface area and nano-scale dimensions [18]. They have shown a preference in addressing several challenges linked to traditional EOR techniques. Furthermore, surfactants are often incorporated into nanoparticle solutions to improve their stability. Surfactant-coated

nano-particles are functionalized nano-particles that include a nanoscale component with surface-active groups designed to perform specific functions, such as adsorbing at the oil-water interface to alter properties like wettability and IFT [9]. In recent decades, polymer nano-particles have been widely used in various fields, ranging from conductive materials to environmental remediation and medical applications [19]. Their low toxicity and high biocompatibility [20] make them suitable for reservoir environments. Additionally, the surface of polymer nano-particles can be readily functionalized, particularly with carboxyl groups [21], allowing for modifications to improve their resistance to temperature and salinity. These properties make polymer nano-particles an excellent choice for use as carriers. Nano-particles with polymer chains grafted to the surface known as polymer-coated nano-particles are an emerging class of materials that may be superior to nano-particles for enhanced oil recovery due to improved solubility and stability, greater stabilization of foams and emulsions, and more facile transport via porous medium [22]. Generally, the combination of nano-particles with polymers or surfactants offers several advantages, including an increase in the viscosity of the displacing fluid [[23], [24], [25], [26]], enhanced emulsion stability [[27], [28], [29]], a reduction in surfactant loss within oil reservoir rocks [[30], [31], [32], [33]], and a decrease in IFT [[34], [35], [36], [37], [38]]. Various works have indicated that the incorporation of nano-particles helps the decrease of the sensitivity of polymer and surfactant flooding to challenging environmental conditions [[39], [40], [41], [42]]. One notable benefit of using nano-particles to recover trapped oil is their ability to generate disjoining pressure. This mechanism involves nano-particles to create pressure at the oil-water interface, which alters wettability and facilitates the movement of oil away from the rock surface [43,44]. Moreover, chemical additives can increase the mobility ratio of the injected fluids compared to those in the reservoir by modifying the viscosity of the injected fluid and the relative permeability of the fluids [45]. Despite extensive research on individual EOR methods, studies often focus on a single type of chemical for oil recovery. In contrast, nano-composites, which consist of two or more materials, can synergistically enhance achieving optimal performance [46]. The role of nano-composites in optimizing IFT and wettability in oil reservoirs contributes to improved oil recovery

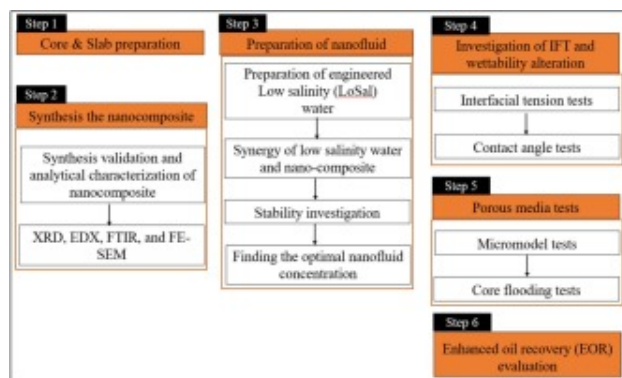
[47]. Khaksar-Manshad et al. [[48], [49], [50], [51]] found that polymer-coated nano-particles are excellent candidates for reducing IFT and shifting the system's wettability towards WW conditions. Nano-silica is notably hydrophilic due to the high density of hydroxyl groups on its surface, making it effective for use in oil fields. However, a challenge associated with nano-particles is their instability in aqueous environments; this instability can be mitigated by incorporating polymers and surfactants to enhance their stability. This research focuses on the injection of a smart nano-fluid containing a newly developed nano-composite to evaluate EOR parameters using an innovative approach. In this regard, tests such as stability, IFT, wettability alteration, micromodel, and core flooding in the presence of nano-fluids were investigated. Generally, in oil reservoirs, this synergistic combination helps compensate for individual limitations, improving the overall efficiency of hybrid EOR methods.

2. Materials and method

2.1. Materials

Carboxymethyl cellulose, Tetra n-Butyl Orthotitanate, tetraethyl orthosilicate, K-laurate, sodium dodecyl sulfate, monopropylene glycol, 2-Propanol(l), Potassium hydroxide, Ethanol, Acetone, Ammonium persulfate, itaconic acid, Acryl amid, N, N'-Methylene bisacrylamide, Cadmium Chloride, and tween 20 from Merk, were used with laboratory purity (>99.9%). All other chemicals and solvents used in this work were in analytical grade. In this study, carbonate rock sample was used for rock and fluid tests. In this regard, plug has porosity (17.3%), length (5.445 cm), dry weight (143.11 d), Swi (26.5%), diameter (3.805 cm), and permeability (4.31 md). The crude oil used in these experiments was prepared from an oil field in western Iran. The density, API, and viscosity of the oil are 0.9463 g/cm³, 18, and 5.12 cP, respectively. Also, the results of SARA analysis are used to better characterize crude oil as saturates (57.83%), aromatics (24.88%), resins (13.85%), and asphaltenes (3.44%). In this regard, SARA analysis has been performed based on ASTM D2007-91, which is based on clay-gel adsorption chromatography [52]. The process

begins with asphaltenes removal, where the sample is treated with a non-polar solvent (n-heptane), causing the asphaltenes to precipitate out, which are then filtered and weighed separately. The remaining deasphalted oil (maltenes) undergoes chromatographic separation in a glass column packed with two distinct adsorbent layers: an upper layer of activated attapulgus clay to trap polar resins and a lower layer of silica gel to separate saturates from aromatics. The sample is dissolved in a solvent (toluene) and introduced into the column, where sequential elution is performed using solvents of increasing polarity. First, n-heptane elutes the non-polar saturates, which pass through both adsorbents with minimal interaction. Next, toluene is used to desorb the moderately polar aromatics from the silica gel, while the clay retains the more polar resins. Finally, a strong polar solvent displaces the resins from the clay layer. Each fraction is collected, evaporated to remove solvents, and weighed to determine its proportion in the original sample. On the other hand, various salts including (sodium chloride, potassium chloride, calcium chloride dihydrate, magnesium chloride hexahydrate, sodium sulfate, sodium bicarbonate) from Dr. Mojallaly Company, Iran (with laboratory purity above 99%) were used to make nano-fluids. In this study, the overall experimental procedure flow chart is shown in Fig. 1.



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Fig. 1. Flow chart of the experimental steps in this research.

2.2. Synthesis and nano-fluid preparation in LoSal water and its characterization

2.2.1. NC synthesis

The synthesis of the nano-composite was carried out by the so-called impregnation method. This means that titanium dioxide and silicanano-particles were each synthesized individually by the sol-gel method and finally combined with a certain ratio. In this regard, the procedure for obtaining silica nano-particles using the sol-gel procedure was based on Zulfikar et al. [53] and Chaturvedi and Sharma [54]. Also, the procedure for obtaining silica nano-particles using the sol-gel procedure was based on Giampiccolo et al. [55]. It is worth noting that metal salts such as nitrate and metal alkoxides were used to synthesize the nano-particles. Metal alkoxides are a group of organometallic compounds that contain an OH base attached to a metal or metal-like element. Their general chemical formula is $M(OR)_n$. For the synthesis of titanium dioxide nano-particles, the precursor Tetra Butyl orthotitanate (TBOT) (a type of metal alkoxide) was used and proceeded via the acidic route. For the synthesis of silica nano-particles, the precursor tetraethyl orthosilicate was used and proceeded from the basic route. The result of the synthesis of nano-particles by the inoculation method is a titanium dioxide/silica nano-composite. Then, it was fluidized with different fluids; (two-step method).

2.2.1.1. Synthesis of titanium dioxide nano-particles by sol-gel method (acid hydrolysis)

To synthesize titanium dioxide, the TBOT precursor was employed, and proceeded through the acidic route. For this purpose, first, 21 g of TBOT was weighed and then dissolved in a mixture of 2-propanol and ethanol. The solution was stirred for about 30min at room temperature. Then, a small amount of ethylene diamine was added to the solution (to stimulate the system and prepare it for acidic hydrolysis), and was stirred again for 45min. In fact, using ethylene diamine as a base prevents the precipitate and hardening. Next, 3.5ml of 65% nitric acid was weighed using a graduated cylinder. To

obtain an aqueous solution, 12cc of ethanol was added to 160cc of water. Next, nitric acid was added to the ethanol solution in water. The resulting solution, i.e. water + nitric acid + ethanol, was added drop wise to the above solution by titration. It should be noted that a slow titration operation made an appropriate precipitate, and finally a good reaction. Then, the temperature of the system was slowly increased from 50 to 80°C. Gradually, the color of the mixture changed to milky and a gel state was obtained. After the completion of the titration, pH=3 was reported through a pH paper. To perform the reflux operation, the solution was stirred at 80°C for 3h. Finally, to modify the structure, better two-phase formation, and the formation of the precipitate, the solution was placed in the refrigerator at 5°C for one day. It is worth noting that the cooling helped the two-phase formation and caused the precipitate to settle well. The nano-fluid was placed at room temperature for about an hour. Then, the nano-fluid was placed on a heater stirrer in order to evaporate its solvent. Then, the obtained gel is poured in a crucible, and then in an oven at 80°C. The powder obtained from the gel is placed in an oven at 260°C for 7h. The titanium dioxide powder is placed in an oven again at 730°C for 6h to obtain titanium dioxide powder nano-particles through a calcination process.

2.2.1.2. Synthesis of silica nano-particles by sol-gel method (base hydrolysis)

In first, amount of 25g of tetraethyl orthosilicate in water was dissolved. Then it was placed on a stirrer at 35°C. Subsequently, 1.5g of sodium dodecyl sulfate, a type of surfactant, was added to the solution. The beaker containing the fluids was covered with foil, and adjusts the reaction temperature to 35°C with stirring for 40min. Moreover, solution containing 6.5g KOH (s)+110cc water (l)+60cc 2-Propanol (l) as a basic hydrolyzing solution was utilized. It was titrated through a decanter in the esterifying solution. After some time, the pH of the solution was measured as 11. Then, the reflux operation was performed for 3h. In the reflux stage, the particles become finer, bonds are established, and the solution gradually becomes cloudy. A precipitate is formed in the hydrolyzing stage. Then, after the reflux operation, the mixture was placed in a refrigerator at temperature of 5°C for making two-phase formation, properly. Then, the silica solution was put in an oven at 110°C for separating the solvent and solute, which is

called the hydrothermal process [23]. Finally, again, it was placed in the oven for 7 h and temperature was gradually increased to 650°C. The temperature was increased to obtain the desired product, which are silica powder nano-particles through a calcination process.

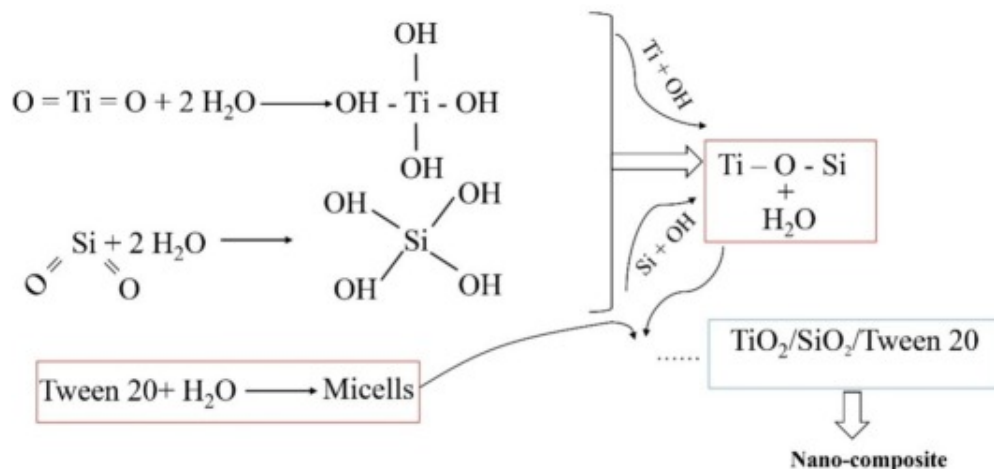
2.2.1.3. Synthesis of titanium dioxide/silica nano-composite by inoculation method

In order to synthesize titanium dioxide/silica nano-composite, the synthesized titanium dioxide and silica nano-particles were determined in a ratio of 20 to 80wt%, poured into a solution of water and isopropanol. Next, surfactants of K-laurate and monopropylene glycol were added to the mixture and refluxed for 5 h. Then it was placed in an oven at temperature of 90°C for one day. The powder was easily separated from the solution and settles. This is absolutely correct because silica and titanium dioxide do not dissolve in water or alcohol. In order to adjoin the materials, they must be mixed and refluxed in a solution for several hours so that the reaction and bonding can take place during the time. The titanium dioxide/silica complex solution was taken from the oven and its solvent was evaporated by a stirrer. Finally, it was washed with water and alcohol, and placed in a refrigerator at 5°C. The titanium dioxide/silica nano-composite was removed from the refrigerator. The solvent of the titanium dioxide/silica nano-composite was separated by a pipette and then placed in a crucible. It was put in a furnace at a temperature of 650°C. Finally, the titanium dioxide/silica powder nano-composite was obtained.

2.2.1.4. Modification of titanium dioxide/silica nano-composite with tween 20

In this section, to further disperse the nano-composite in the water-based fluid, Tween 20 used to modify the surface of the nano-composite. For this aim, a new formulation was used to make titanium dioxide/silica/Tween 20 nano-fluid. First, 3.5 g of the nano-composite was poured into a solution containing distilled water and isopropanol, and then small amounts of Tween 20 were added to it. The obtained solution was stirred on a heater stirrer at a temperature of 60°C. Then, 20 ml of acetone and some carboxymethyl cellulose were added to the solution. Finally, the new nanocomposite was synthesized. In this regard, the formation of a TiO₂-SiO₂-Tween 20 nanocomposite typically involves a

sol-gel process where titanium and silicon precursors hydrolyze and condense in the presence of Tween 20 (a nonionic surfactant). In this study, an outline of nanocomposite manufacturing and how the reactions perform in Fig. 2 was reported.



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Fig. 2. Chemical reaction scheme to form NC.

2.2.2. Nano-fluid preparation

In this study, a digital heater stirrer manufactured by Heidolf was used to homogenize and stir various fluids at a specific temperature. The stirrer rotation speed can be changed digitally up to 1300rpm and the heating plate temperature can be changed from 25 to 300°C. In general, one of the parameters affecting the performance of nano-particles is the type of base fluid. Nano-fluids were prepared by dispersing NC containing surfactant in the LoSal water. The composition of LoSal water includes NaCl (12.788 Wt%), KCl (0.559 Wt%), CaCl₂·2H₂O (0.882 Wt%), MgCl₂·6H₂O (5.997 Wt%), Na₂SO₄ (3.409 Wt%), and NaHCO₃ (0.168 Wt%). The composition of ions in the fluid can influence the efficiency of oil recovery from reservoirs. LoSal water injection, which contains reactive ions, achieves a higher oil recovery rate than deionized water injection because it forms electrostatic

bonds between ions and negatively charged fatty acids in the oil. In the following, the formulation was subjected for sonication for 6h in an ultrasonic liquid processor. Thus, the newly prepared nano-fluids (smart nano-fluids) were prepared.

2.2.3. Characterization of Nano-fluids

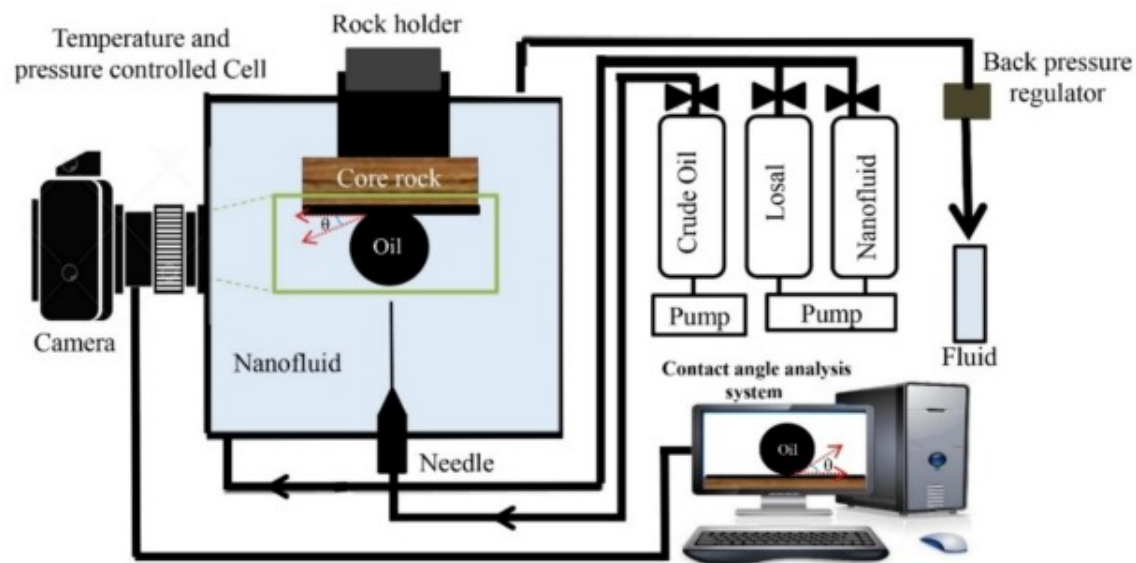
Since obtaining live oil from a reservoir is very costly in terms of economic and experimental conditions, in this study, dead oil from one of the reservoirs in western Iran was used after filtering and passing through 2.5 μ m paper. The combination of smart nano-fluid shows great promise in addressing wettability challenges and improving oil recovery rates in reservoirs. The smart nano-fluid prepared in this paper, which consists of surfactants/nano-particles/LoSal water, exhibits remarkable physicochemical properties that are not successful with individual components alone. In this regard, in the following sections, the analysis of the nano-composite properties is presented.

2.3. Evaluation of stability and contact angle measurement

ζ potential measurement is often used to assess the stability of suspensions. In this study, ζ potential analysis was done on various concentrations of nano-fluid to identify the most stable nano-fluid. The ζ potential, consisting of electrostatic repulsion force and van der Waals force, is a comprehensive reflection index [56]. In general, electrostatic repulsion exists among nano-particles [57]. Greater electrostatic repulsion correlates with improved stability of nano-fluids [58,59]. In this study, in order to determine the stability of colloidal dispersions, eight nanofluid concentrations (100, 200, 300, 400, 500, 600, 700, and 800ppm) were investigated with LoSal water. To perform the ζ potential test, half of the carbonate sample was passed through a 5 μ m sieve after being pulverized by a ball mill to ensure the size and uniformity of the powder. The other half of the core was cut into four parts of equal thickness and, after polishing their surfaces, was used in the contact angle test. For further and detailed investigation the effect of the nano-fluids on wettability changes, in addition to ζ potential, the contact angle was also applied. In the ζ potential test, the powder from the core rock sample was poured into a small container

and placed in contact with the dead oil of the reservoir. Similarly, the carbonate slices were also placed in contact with the dead oil of the reservoir. Then, the powder and core rock slices were placed in the aging cell to bring their conditions closer to the reservoir conditions. The cylinder and its contents remained at a temperature of 90°C and a pressure of 550psi for 37 day. After the aging period, 0.01 of the rock powder was placed in contact with the nano-fluid sample. In order to achieve greater uniformity, the homogenization step was performed by an ultrasonic device to reach equilibrium conditions, after 48h, the ζ potential of each sample was measured using a ζ potential device under ambient conditions. Also, observations of the ionic forms of the solutions were done using Inola Cond7110 conductivity measurements. In this regard, the electrical conductivity of the fluids was measured to find the stability of smart nano-fluid. If the electrical conductivity of the nano-fluids exhibits a small deviation, the nano-fluid will be stable. In the contact angle test, the core slices that had completed their aging period, were placed in a contact angle measuring device under ambient conditions and the surroundings of the samples were filled separately with nano-fluid. The oil sample was injected into the system from below the device and the drop was placed under the sample in such the OW wettability of the carbonate sample was maintained. Over time, changes in the contact angle between the oil and the rock sample were recorded in the presence of different concentrations. Depending on the type of the nano-fluid, the experiment continued for 24 to 47h (until no further change in the contact angle was observed), and photographs were taken of the samples at specific intervals. The photographs were transferred to a computer and the angles related to the drop and the rock surface, were measured in the presence of the nano-fluid through Image J software.

Fig. 3 depicts a schematic of the contact angle measurement.



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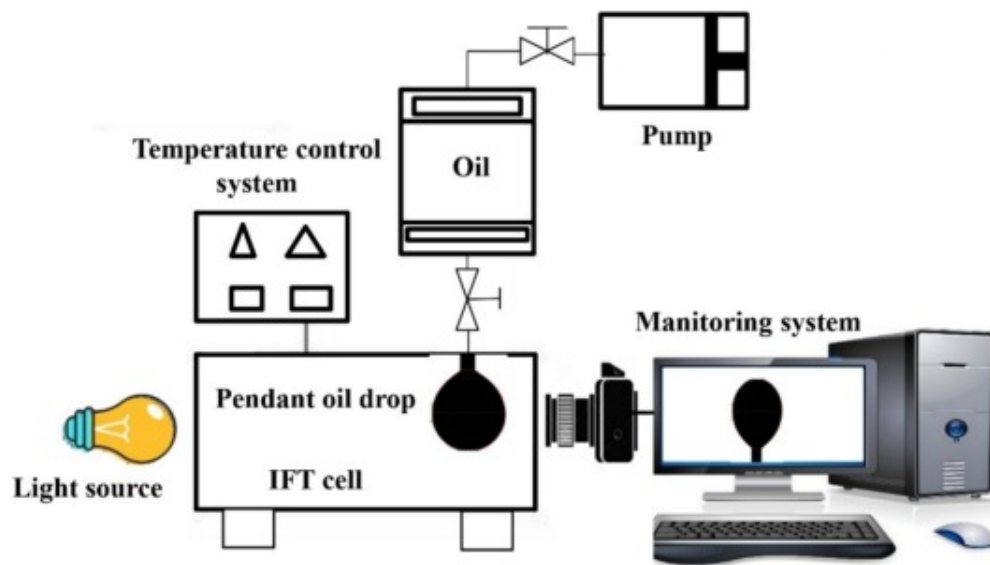
Fig. 3. Schematic diagram of the contact angle measurement system.

2.4. IFTstudy

IFT is recognized as a key parameter for assessing fluid distribution and movement within porous media. Furthermore, IFT plays a significant role in EOR through nano-fluid flooding. Therefore, precise measurement of the IFT between oil and water is essential for EOR applications [56]. In this regard, the pendant drop method is one of the most straightforward techniques for measuring IFT. In this regard, Fig. 4 presents a schematic representation of the IFT measurement process. The accuracy of this method relies on the cleanliness of the apparatus; thus, the device's cell must be thoroughly cleaned with toluene, water, and acetone in that order prior to IFT measurement. The nano-fluid is then prepared in various concentrations of 100–800ppm. Subsequently, the solutions are prepared and introduced into the main chamber. Then oil is injected into this chamber to establish a mechanical equilibrium between the two phases. Based on Fig. 4, in order to

measure interfacial tension, a U-shaped syringe is needed to place the oil droplet at its tip. Thus, the tip of the needle on which the oil drop is placed, is put on the chamber and then image of the drop is recorded using the camera. The device captures images of the oil droplet within the solution, and IFT is calculated through image analysis. In this regard, to assess the droplet characteristics and ultimately calculate the IFT, Eq. (1) is introduced as follows [60].

$$\sigma = \frac{\Delta\rho g D}{H} \quad (1)$$



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Fig. 4. Schematic diagram of the IFT measurement system.

In Eq. (1), σ , D , d , ρ_A , ρ_B , $\Delta\rho$, and H represent the interfacial tension (IFT, mN/m) between the two fluids, the maximum diameter of the pendant drop (μm), the diameter at a point located a distance D from the drop apex (μm), the density of water, the density of oil, the density difference between the aqueous phase and the oil droplet, and the shape

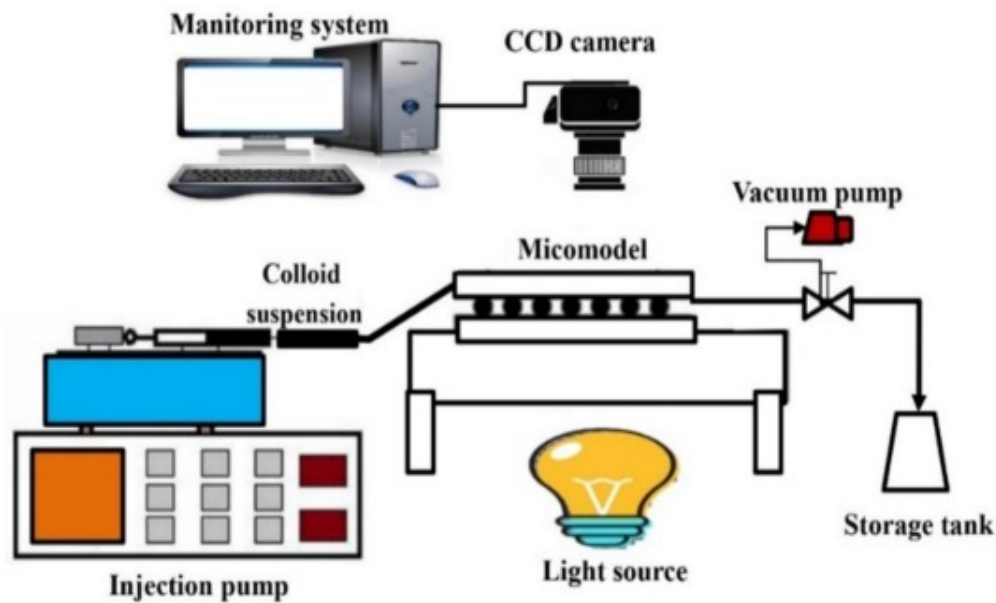
parameter, respectively. The shape parameter is generally a function of the shape factor ($H=S=d/D$).

2.5. Displacement test

The micromodel is a device designed to demonstrate fluid flow within porous media. To create a glass micromodel, two sheets of glass were selected, and a porous media pattern was etched onto them using a laser. After completing the micromodel, it was connected to a pump with a high-precision of 10^{-4} cc/h. The micromodel has a permeability of 13 md, porosity of 8%, a length of 7.7 cm, a width of 5.7 cm, and a depth of 0.27 cm. On the other hand, schematic representation can be seen in [Fig. 5](#). A high-quality camera captured images of the micromodel, which were utilized to calculate the oil recovery. One of the key benefits of using a glass micromodel is its transparency, which facilitates the study of multiphase flow within pore structures through microscopy. In this research, the micromodel was rendered OW using an organochlorosilane (2wt%) solution in toluene (98wt%) [[59,61](#)]. For each experiment, the micromodel was thoroughly saturated with crude oil. Then, it was balanced to negate the effects of gravity on fluid movement. Subsequently, the oil-saturated micromodel was flooded with the LoSal water for approximately five pore volumes until no oil was produced. The micromodel was flooded with nano-fluids at various concentrations (100, 200, 300, 400, 500, 600, 700, and 800ppm) for a similar volume of approximately five pore volumes, akin to the LoSal water flooding process. The experiments using nano-fluids were compared against those using the LoSal water. In all experiments, the injection rate was maintained at 0.03 ml/h. To assess EOR in carbonate core plug, core flood tests were conducted to investigate the effects of nano-fluids. Generally, to prevent contamination, the carbonate core plug underwent soxhlet extraction for 72h, followed by washing with methanol. The core plug was rendered OW by immersing them in crude oil for four weeks. The core flood tests were carried out at room temperature, with an injection rate of 0.2 ml/min for both deionized water and nano-fluids. The LoSal water flooding continued for approximately two pore volumes, and once oil production ceased during this phase, nano-fluid flooding commenced, which also lasted for around five pore volumes. Then, cumulative oil

recovery was calculated, where “oil in place” refers to the volume of oil present. To determine this volume, first the LoSal water was injected, followed by crude oil until no LoSal water was produced. Due to the variability in core plug characteristics, it is reasonable to calculate the ratio of recovered oil to total oil recovery to quantify the amount of extracted oil. Generally, cumulative oil recovery was gained based on Eq. (2).

$$\text{Cumulative Oil Recovery} = \frac{\text{Produced Oil}}{\text{Oil in place}} \times 100 \quad (2)$$



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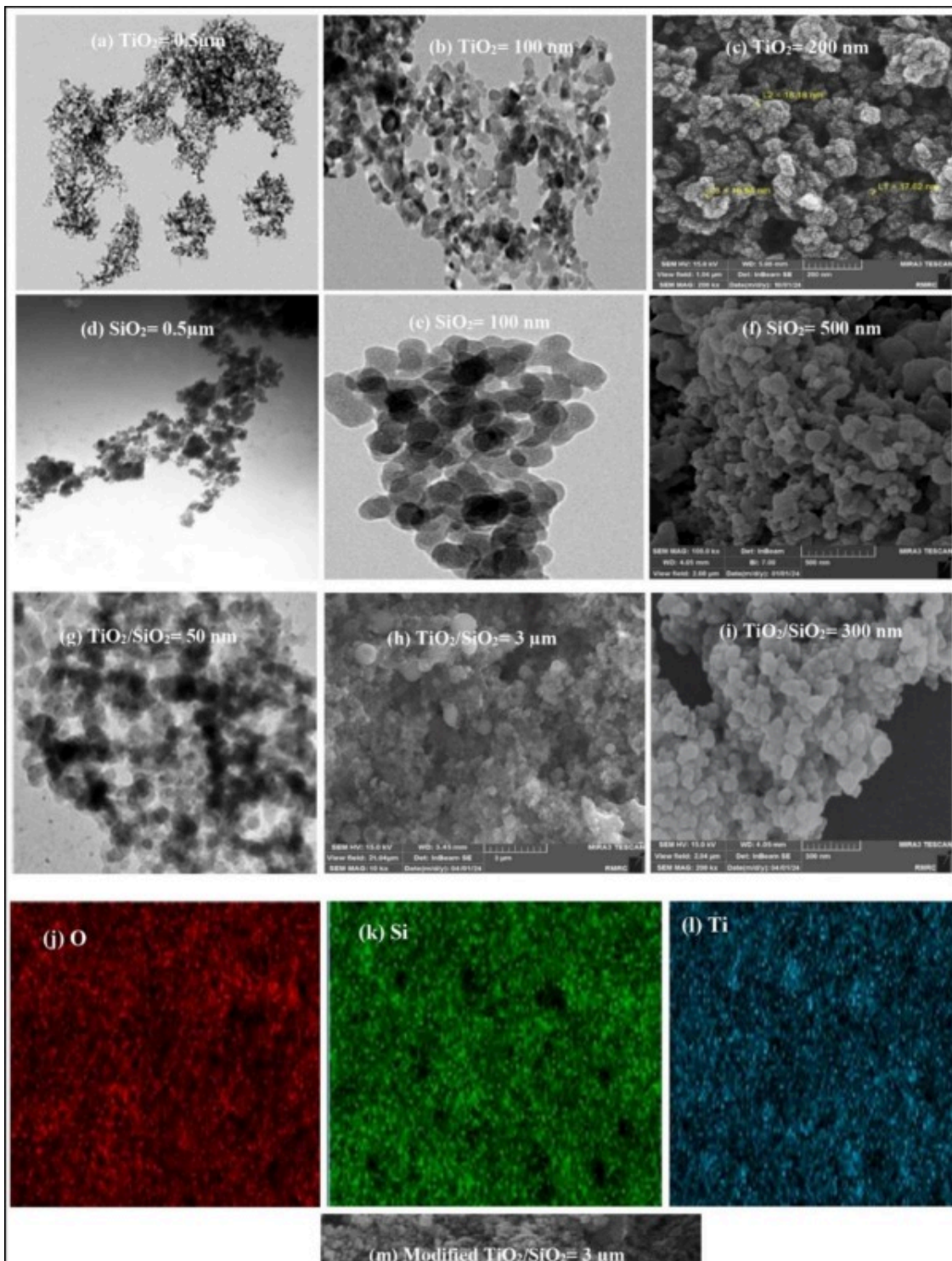
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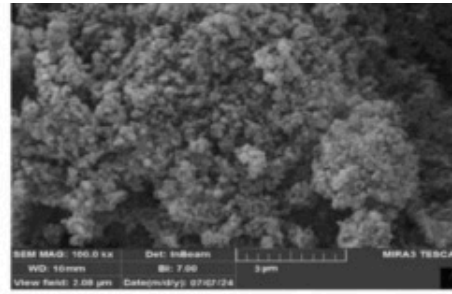
Fig. 5. Schematic diagram of the micromodel experiment set-up.

3. Results and discussion

3.1. Characterization of the nano-particle and nano-composite

FE-SEM and TEM microscopy techniques were employed to analyze the morphology of the synthesized compounds, as illustrated in Fig. 6 (a–m). Fig. 5 (a, b, d, e, g) displays TEM images of titanium dioxide and silica nanoparticles at varying sizes, while Fig. 5 (c, f, h, i) presents FE-SEM images of titanium dioxide/silica NC at different scales. The FE-SEM image of the novel NC is shown in Fig. 6 (m). Additionally, energy-dispersive X-ray (EDX) elemental mapping confirmed the distribution of oxygen (O), silicon (Si), and titanium (Ti), as depicted in Fig. 6 (j–l). The modified NC is illustrated in Fig. 6 (m). Furthermore, EDX spectroscopy, a widely used technique for elemental analysis, was utilized to examine the composition of the titanium dioxide/silica NC. In this regard, Fig. 7 shows the EDX spectrum of the titanium dioxide and silica NC.

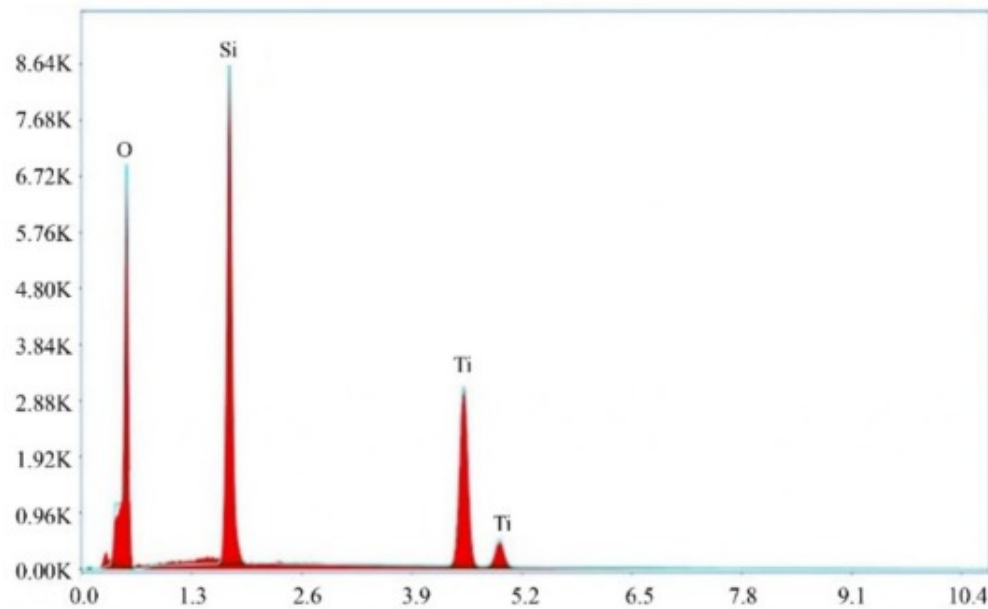




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Fig. 6. TEM and FE-SEM images of TiO_2 , SiO_2 , novel nano-composites, and energy dispersive X-ray (EDX) spectroscopy.



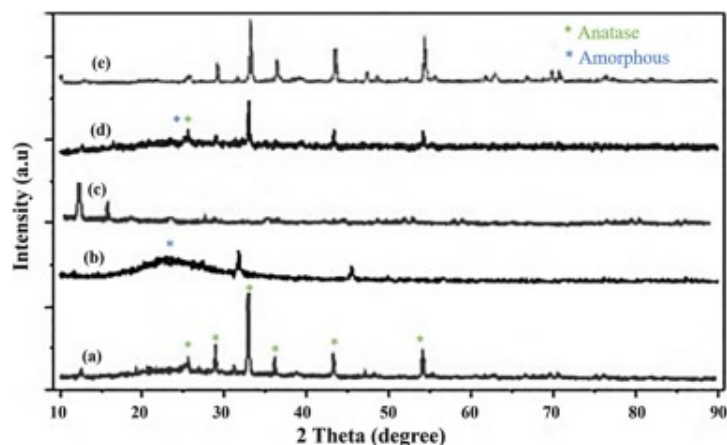
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Fig. 7. Elemental analysis by EDX spectrometry for titanium dioxide/silica.

Moreover, the characteristics of the newly synthesized NC were investigated using XRD analysis. In this regard, Fig. 8 displays the XRD for titanium dioxide, silica, Tween 20, titanium dioxide/silica, and the novel NC. Based on Fig. 8(a), the titanium dioxide pattern reveals distinct peaks, which correspond to the anatase phase of titanium dioxide (JCPDS No. 0075–1537), confirming that anatase is the predominant phase in titanium dioxide [62,63]. These peaks are at 25.88, 29.15, 33.12, 36.33, 43.50, and 54.33° [62,63]. In Fig. 8 (b), the diffractogram for pure silica shows a relatively smooth pattern indicative of an amorphous phase. Based on Fig. 8 (c), the XRD pattern of Tween 20 shows a bulge centered at about 15 to 25°, which is characteristic of disordered, non-crystalline materials. The titanium dioxide/silica NC (Fig. 8 (d)) exhibits a similar pattern to that of titanium dioxide (Fig. 8 (a)) but with reduced peak intensity, indicating bond formation within the NC. This bond is in the form of titanium dioxide-O-silica (peak at 23–24°) [62,63]. The XRD analysis reveals that titanium dioxide maintains strong crystallinity, evidenced by the distinct diffraction peaks matching the anatase phase (Fig. 8a). These findings demonstrate that titanium dioxide (based on Fig. 8 (d)) primarily crystallizes in the anatase structure, further validating the NC phase composition. Also, the titanium dioxide peaks in 8 (e) are broader compared to 8 (a), indicating a reduction in crystallite size due to the presence of silica/Tween 20. By adding Tween 20 to the nanocomposite, a new structure of TiO₂/SiO₂/Tween 20 NC was confirmed. On the other hand, FT-IR analysis was conducted to identify the functional groups in the newly synthesized NC. In this regard, Fig. 9 presents the FT-IR spectra for silica, titanium dioxide, titanium dioxide/silica and the novel NC. In Fig. 9 (a), the band around 500–800 cm⁻¹ represents the Ti-O-Ti stretching vibrations. In Fig. 9 (b), the band at about 800 cm⁻¹ (Si-O-Si symmetric stretching) and about 470 cm⁻¹ (Si-O-Si bending) are shown [61,62]. Fig. 9 (c) shows peaks at around 2900 cm⁻¹ (C—H stretching), around 1740 cm⁻¹ (C=O of ester groups) and around 1100 cm⁻¹ (C-O-C ether bonds). Also, Fig. 9 (d) shows the overlapping peaks of Ti-O-Ti (around 500–800 cm⁻¹) and Si-O-Si (around 1100 cm⁻¹). Furthermore, in Fig. 9(e), peak 3476 cm⁻¹ indicates the presence of O-H-bonded, a signal that is due to the presence

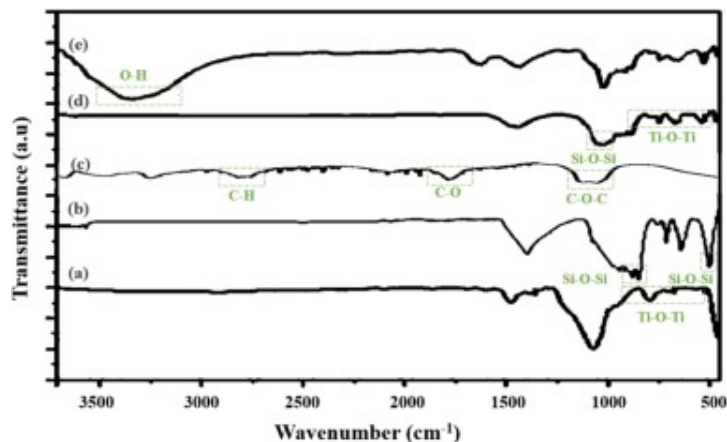
of Tween 20 in the new NC composition. Due to the presence of O—H bonds in the synthetic composition, a water-wet approach was developed for the core rock. Generally, the titanium dioxide/silica/Tween 20 NC (Fig. 9 (d)) exhibited further attenuation of the titanium dioxide peaks, likely due to the organic coating. Meanwhile, the silica signal persisted, though it may have been partially obscured by organic components.



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Fig. 8. XRD patterns (a)TiO₂, (b) SiO₂, (c) Tween 20, (d) TiO₂/SiO₂, and (e) novel NC samples.



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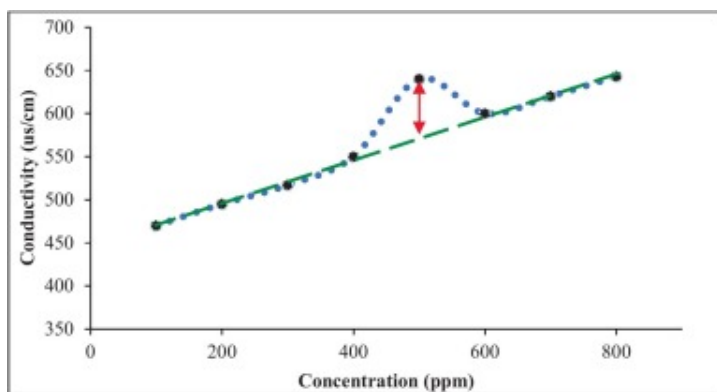
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Fig. 9. FT-IR patterns (a)TiO₂, (b) SiO₂, (c) Tween 20, (d) TiO₂/SiO₂, and (e) novel NC samples.

3.2. Stability assessment and measurement of contact angle and IFT

The integration of nanoparticles and surfactants can modify the wetting characteristics of a reservoir, shifting it towards a WW condition. Typically, the small size of nano-particles allows them to navigate through pore throats in standard reservoirs, reaching residual oil that conventional enhanced oil recovery methods cannot access. In this research, the electrical conductivity of prepared synthetic mineral water with varying concentrations of nano-fluid (100, 200, 300, 400, 500, 600, 700, and 800ppm) was assessed. As illustrated in Fig. 10, a linear relationship between conductivity and concentration was observed represented by a green line. It indicates that the conductivity of the solution increases as the concentration reaches 500ppm. Notably, an unusual behavior was detected at this concentration, highlighted by a red anomaly line. This nanoscale incompatibility arises from the effective electrical conductivity of colloidal suspensions in fluids, reflecting a complex interplay of factors such as ion release, volume fraction, electric double layer characteristics, and other physicochemical properties that standard

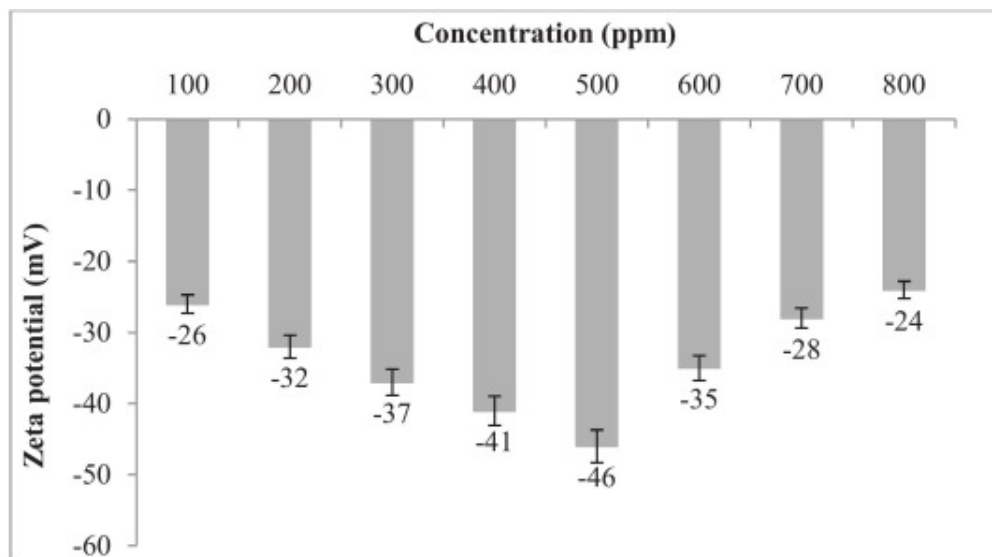
models fail to accurately depict [5,46]. The observed increase in conductivity at 500ppm is attributed to the electrical interactions between the two layers and the influence of the net charge of solid particles, leading to stronger interactions at this concentration. Consequently, these mechanisms produced an abnormal signal in the conductance relative to concentration at 500ppm. The ζ potential results are displayed in Fig. 11, revealing that among the nano-fluids tested, the 800ppm of the nano-fluid exhibited the highest ζ potential value of -24mV , while the lowest (most negative) value was recorded for the 500ppm of the nano-fluid at -46mV . This indicates that the nano-fluid with a concentration of 500ppm has long-term stability. In other words, a more negative ζ potential is an expression of the greater power of the nano-fluids to separate the carboxyl group bound to the surface, which results in an increased oil recovery rate. Then, the colloidal stability of the nano-fluid was assessed by analyzing the particle size distribution via dynamic light scattering (DLS). After preparation, dispersion stability tests were performed at both room temperature (25°C) and elevated temperature (80°C). Over a seven-day period, nanoparticle aggregation and precipitation were monitored visually. The nano-fluid with a concentration of 500ppm exhibited the highest stability, as confirmed by visual inspection. DLS analysis further supported this finding, revealing that the nano-fluid retained a consistent particle size of 100nm over the seven-day period, demonstrating excellent long-term stability under both ambient and high-temperature conditions (Fig. 12).



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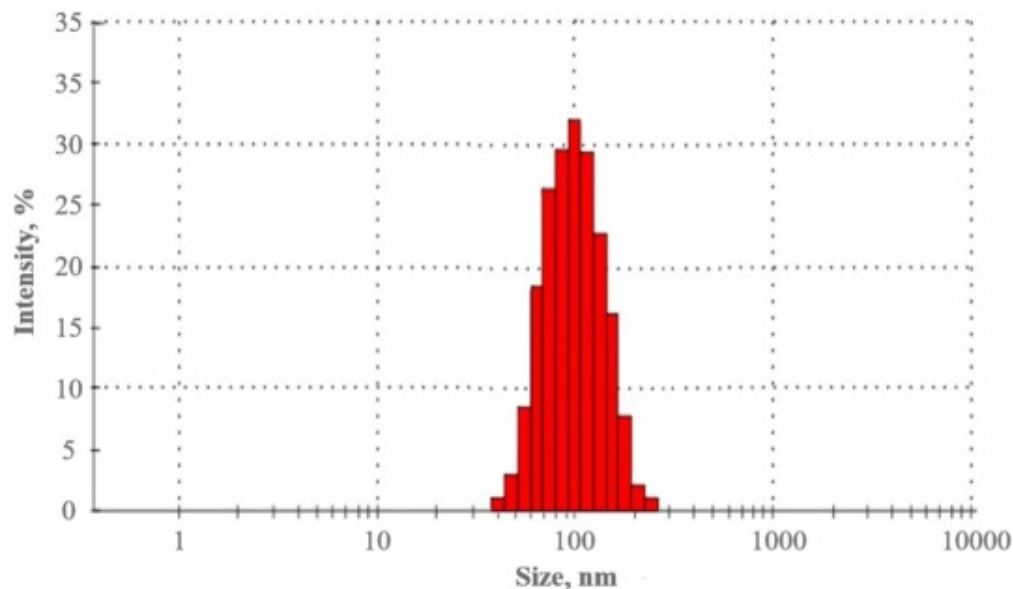
Fig. 10. Conductivity values of the nano-fluidssolution.



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Fig. 11. The ζ potential of the nano-fluids.



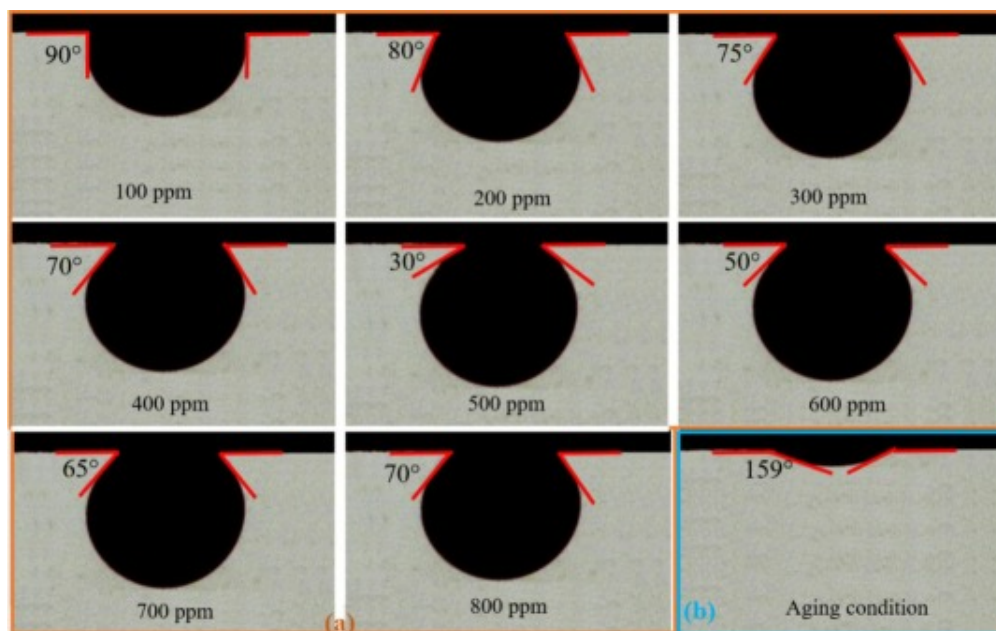
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Fig. 12. DLS results for NC.

Also, to measure the contact angle, the conventional sessile drop method was used. In this approach, the testing procedure began by saturating a rock sample with oil. The contact angle was then measured using Losal water, which showed an initial oil contact angle of 159° for the aged core sample. Next, the oil-saturated rock was treated with different nano-fluids, and the contact angles for all concentrations were measured using an IFT700–32175 instrument. Fig. 13 shows the schematic of the changes in the contact angle of the oil sample and the core rock in the presence of nano-fluids. Also, as can be seen in Fig. 14, with increasing concentration to 500ppm, the final contact angle between the rock and the oil droplet reduces further. So that the final contact angle of the oil in the presence of nano-fluid (500ppm) reaches 30°, which indicates a significant change in the wettability of the rock towards hydrophilicity. In fact, the interaction of the nano-fluid, their ability to move from the dissolved mass to the rock surface increases due to the attraction of opposite charges. As a result, this process leads to the establishment of ionic

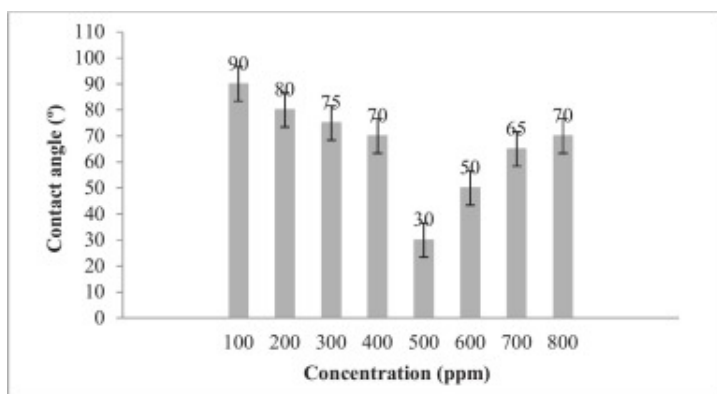
bonds between calcium ions present on the surface and free sulfate ions in the dissolved mass, which causes the release of carboxylic groups. As the carboxyl groups are released, the adhesion of the rock to the oil phase decreases and the tendency of the oil droplet to separate from the carbonate surface increases, which is reflected in a decrease in the contact angle. The observed changes in Fig. 14 indicate that the ability of the nano-fluid to change the wettability of the carbonate reservoir rock is directly related to the trend of the ζ potential test results, and the results of both tests confirm each other. As a result, the 500ppm solution exhibited a significant negative charge concentration, as indicated by the electrical conductivity and ζ potential tests. Generally, in this section, each experiment was done three times to check repeatability. The findings demonstrated that the nano-fluid diminishes oil adsorption on rock surfaces by generating a structural disjoining force (film tension) that forms a wedge film structure between the oil and the rock. This smart nano-fluid regulates the disjoining pressure on the rock surfaces, allowing it to coat them effectively. This coating creates a WW layer on the surface. One of the key factors influencing the enhanced oil recovery is the concentration of the smart nano-fluid within the base fluid. The nanoparticles enhance the disjoining pressure, which results in increased Brownian motion due to heightened repulsive forces among the particles [5]. Generally, the results showed that the nano-fluids had a great potential to improve properties of the oil recovery, and interfacial tension especially by adding a nano-composite to the chemical solutions [5]. Based on Fig. 15, the lowest IFT occurred at concentration of 500ppm. Earlier, this concentration was determined as optimum and was confirmed through the other analyses.



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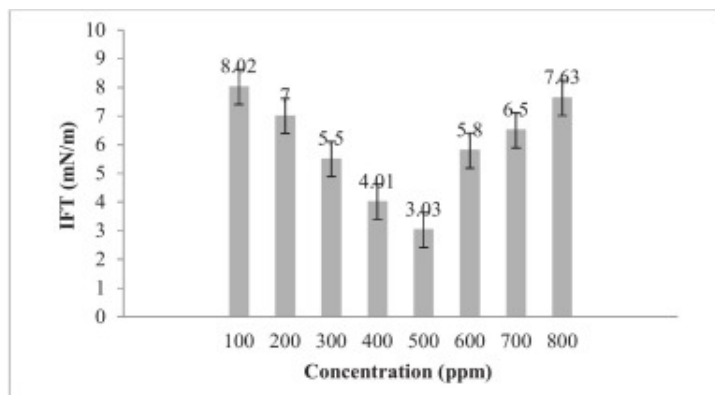
Fig. 13. Schematic view of the contact angle.



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Fig. 14. Contact angle measurements at different concentrations (100, 200, 300, 400, 500, 600, 700, 800ppm) of the nano-fluids.



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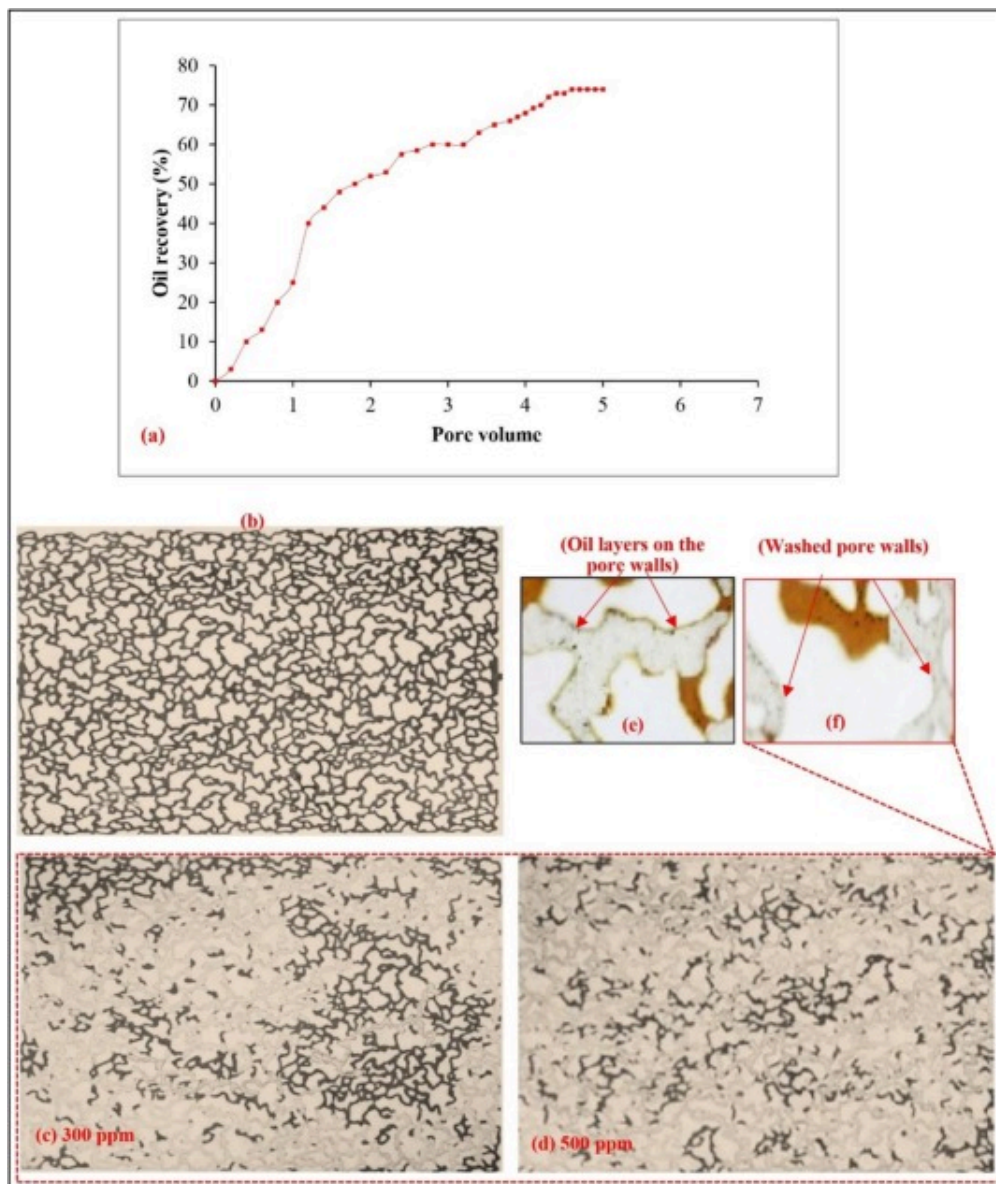
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Fig. 15. IFT measurements at various concentrations (100, 200, 300, 400, 500, 600, 700, 800ppm) of the nano-fluids.

3.3. Micromodel system

Micromodel experiments were conducted to investigate the efficacy of the prepared nano-fluids in EOR, involving various micromodel tests and a repeat trial. Initially, the glass micromodel was saturated with oil, followed by the injection of the LoSal water and the nano-fluids to measure the volume of produced oil. The results of the micromodel tests were analyzed through imaging. Oil recovery was calculated based on the images captured after the injection commenced. In this regard, only 40% of the oil was recovered using LoSal water flooding, while the injection of nano-fluid at a concentration of 500ppm resulted in a 74% oil recovery rate (Fig. 16a). The micromodel experiments demonstrated that the nano-fluid (500ppm) was significantly more effective for oil extraction. Based on the results, given that the pore walls are OW, the movement of oil within the pores is challenging without nanoparticles. Thus, once LoSal water traverses to

the end of the micromodel, oil production ceases. The water tends to follow the most permeable paths due to fingering effects. Fig. 16 (b), depicts the oil-saturated micromodel. In addition, the micromodel with the nano-fluid flooding of 300ppm and 500ppm are shown in Fig. 16(c) and (d), respectively. With a simple comparison, it is obviously observed that a more EOR has occurred at 500ppm which was verified earlier. Additionally, Fig. 16(e) and (f) illustrate the pore walls of the glass micromodel. Analysis of images from various sections of the micromodel indicates that layers of oil adhering to the pore walls have been removed after nano-fluid flooding.



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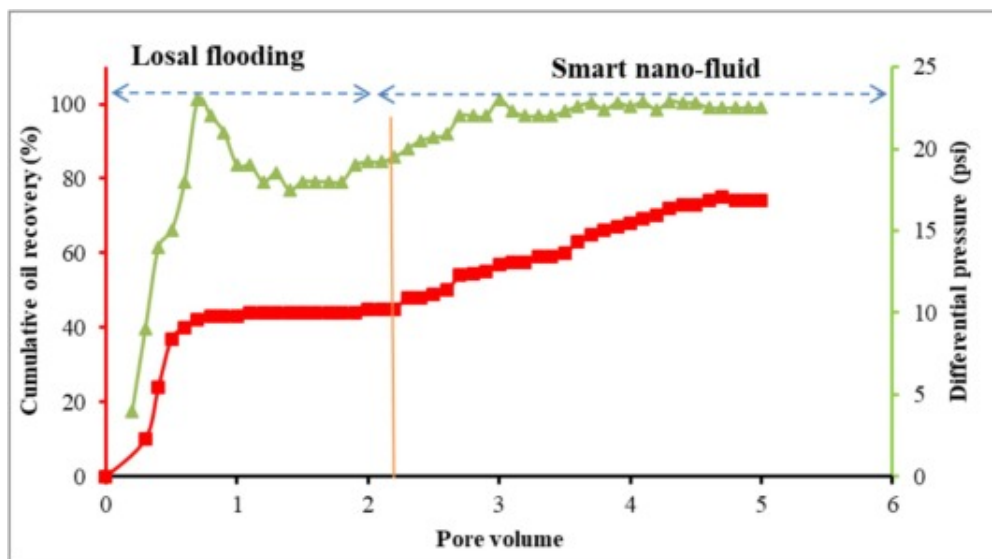
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Fig. 16. Oil recovery after (a) nano-fluid flooding at concentration 500ppm of nano-fluid(micromodel test), (b) Oil-saturated micromodel, (c,d) nano-fluid Flooding (300ppm

&500ppm), (e) the glass micromodel after LoSal water flooding, and (f) the glass micromodel after nano-fluid flooding.

3.4. Coreflood system

In this section, core flood experiments were carried out to evaluate the performance of the prepared nano-fluids in enhanced oil recovery. Fig. 17 depicts oil recovery against injected pore volume obtained from core flood tests using the nano-fluid (500ppm). The results indicate a breakthrough of 45% with LoSal water flooding in the core plug. Following the nano-fluid flooding, oil recovery reached to 74%. This suggests that in the core flood test, the nano-fluid contributed an additional 29% to EOR. Furthermore, Fig. 17 shows the oil recovery for the core sample. Overall, in this study, key findings (IFT, contact angle, and oil recovery data for concentrations of 100–800ppm) are shown in Table 1.



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Fig. 17. Oil recovery for the LoSal flooding and the nano-fluid(500ppm) flooding through a core plug.

Table 1. Key findings (IFT, contact angle, and oil recovery data for concentrations of 100–800ppm).

Nanofluid	Concentration (ppm)	IFT (mN/m)	Contact angle (°)	Oil recovery in the core flooding process (%)	Oil recovery in the micromodel process (%)
TiO ₂ -SiO ₂ - Tween 20	100	8.02	90	18	40
	200	7	80	20	45
	300	5.5	75	24	60
	400	4.01	70	27	64
	500	3.03	30	29	74
	600	5.8	50	23	70
	700	6.5	65	21	67
	800	7.63	70	19	64

4. Mechanism

In this study, the formulated compound showed special efficiency due to its unique properties. Generally, tween 20 used in nano-fluid is a non-ionic surfactant. There is functional group in tween 20 that is a very strong stabilizer and also further reduces IFT. Generally, when nano-fluid is introduced to porous media, the surfactant binds to the structures of acid, resulting in the removal of acid from the rock surface. This process leaves a thin layer of positively charged surfactant on the rock. The presence of nano-particles not only increases the viscosity of the aqueous phase but also decreases the apparent viscosity of oil and improves its rheological properties. Additionally, the positioning of nano-particles at the oil-brine interface, along with their interaction with

surfactants, generates surface pressure that, along with ion exchange, facilitates the detachment of oil from the calcite surface. The results indicated that a combination of nanoparticles, surfactants, and low salinity water achieves optimal synergistic effects, resulting in the lowest contact angle and maximum oil detachment from calcium carbonate surfaces. At a concentration of 500ppm, the strong negative charge causes the nano-fluid to act like an electrolyte. Increasing the concentration of the nano-fluid raises the system's entropy, which is further affected by disjoint pressure. The carbonate core rocks possess positive surface charges, creating a gravitational attraction between the positively charged rock and the highly negatively charged nano-fluid. The electrolytic characteristics of the nano-fluid, along with its surface interactions, position its adsorption as the key mechanism for changing the wettability of the rock. The nano-fluid has a negative surface charge, and when it interacts with water, hydroxyl groups such as Si-OH and Ti-OH are formed. These nano-fluids readily adsorb onto the rock surface, facilitating the formation of additional hydroxyl groups upon contact with water. Also, due to the interaction of OH in the nano-fluid mixture and the rock surface, a certain wettability shift towards water-wet occurs. Consequently, the optimized nano-fluid creates a thin, wedge-shaped layer at the boundary where rock, water, and nano-fluid meet. When the disjoining pressure exceeds the adhesive force binding the oil droplet to the rock surface, the oil detaches. Alterations in wettability result from a combination of chemical and physical processes, along with the influence of electrostatic, van der Waals, and structural surface forces. Additionally, the disjoining pressure near an inner contact line can be influenced by various factors affecting the system's energy and entropy, including structural, electrostatic, and hydrophobic effects. The interaction between hydroxyl group in the nano-fluid and the rock surface leads to a change in hydrophilicity conditions, with the formation of a wedge layer at the interface. When the structural pressure gradient between the oil droplets and the rock increases, oil separation occurs. These wettability changes are due to various interactions, including electrostatic and van der Waals forces. Furthermore, the incorporation of low salinity water ions into the nano-fluid enhances the removal of acids from rock surfaces. In general, divalent ions occupy more space, create a larger surface charge, and lead to a significant reduction in interfacial

tension. Based on the results and the interaction of the smart nano-fluid with rock/fluid, a higher recovery rate is achieved compared to other nano-particles. In general, silica is widely utilized in tertiary oil recovery because of its low cost, high purity, large specific surface area, ease of surface modification, and environmental friendliness. Based on current research, investigating silica and titanium in conjunction with the surfactant Tween 20 represents a significant area for future development of nano-particle-surfactant fluids. In summary, the findings of this research indicate that synergistic nano-fluid can effectively reduce interfacial tension, modify core rock wettability, and boost oil recovery all at the same time. Unlike previous research (Table 2), where a single synthetic nano-fluid rarely achieved all three effects, this approach demonstrates comprehensive performance.

Table 2. Comparing the results of this paper with previous research.

Nano-fluids	Interfacial tension (mN/m)	Contact angle (°)	Incremental oil recovery (%)	Reference
TiO ₂ -SiO ₂ -Tween 20	Reference value to 3.03	Reference value to 30	29%	Current study
SiO ₂ /seawater	–	Reference value to 23.2	–	[60]
SiO ₂ /imidazolium-based ionic liquid ([C12mim][Cl])/gemini surfactant/Tween 80	Reference value to 7.8	Reference value to 20	–	[64]
SiO ₂ /Triton-x100/sodium lauryl sulfate /octylpyridinium chloride	Reference value to 0.1	Reference value to 32.5	–	[65]
SiO ₂ /imidazolium-based IL ([C12mim][Cl], anionic surfactant (AOT))/Tween 80	Reference value to 4	–	–	[66]

Nano-fluids	Interfacial tension (mN/m)	Contact angle (°)	Incremental oil recovery (%)	Reference
Amine/organosiloxane@Al ₂ O ₃ /SiO ₂ /smart water	Reference value to 21	Reference value to 38	7.5	[67]
Smart water/TiO ₂	Reference value to 5.31	Reference value to 80	21	[68]
ZrO ₂ /NaCl and NiO/NaCl	–	Reference value to 38 (ZrO ₂ /NaCl)/ Reference value to 48 (NiO/NaCl)	–	[69]

5. Conclusions

The synthesized NC (TiO₂/SiO₂/Tween 20) was examined through various structural analysis techniques, including TEM, EDX, EDX Mapping, XRD, FT-IR, and FESEM, confirming the success of the synthesis. This study confirmed the successful synthesis of NC through FT-IR analysis. Surface modification enhanced the NC's reactivity, with visible changes evident in the results. Additionally, FE-SEM imaging clearly revealed the structure of the newly developed NC, along with the average particle size distribution. In this regard, in the next step, smart nano-fluid was prepared. To assess the effectiveness of the smart nano-fluid, a series of tests were conducted such as ζ potential, electrical conductivity, contact angle measurements, IFT assessments, core flooding experiments, and micromodel test. Generally, the addition of surfactant was able to significantly improve the stability of the solution. Also, this work demonstrated that the oil recovery

improved through the nano-particle flooding, and in the presence of surfactants in the solution. In general, the stabilization of the nano-particles is sensitive to the ratios of surfactant to nano-particle concentrations. In this study, at a low surfactant-to-nano-particle concentration ratio, the adsorption of surfactants onto the particle surface results in a hydrophilic surface. The findings indicated that the smart nano-fluid (nano-hybrid) presents a good stability in aqueous solutions, with optimal stability observed at a concentration of 500ppm. The nano-hybrid effectively reduced the IFT and altered wettability from OW to WW conditions. The micromodel as a good simulation of reservoir, revealed a clear observation of the oil recovery and the flooding test. Overall, the results showed that core flooding by nano-fluid improves oil recovery by 29%. This study not only elucidates the mechanisms underlying oil recovery during hybrid flooding in low-permeability carbonate rocks but also offers a new reference point for developing innovative CEOR agents. Future research should focus on Investigate surface modification techniques to enhance nano-particle dispersion and prevent aggregation. Also, future studies should focus on determining the optimal parameters and most suitable coating materials for surface modification. By addressing these aspects, the integration of materials (synergistically) into the enhanced oil recovery process can significantly increase oil recovery and contribute to more sustainable energy management.

CRediT authorship contribution statement

Akbar Mohammadidoust: Writing – review & editing, Supervision, Formal analysis, Data curation. **Ehsan Jafarbeigi:** Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] H. Bahraminejad, A. Khaksar-Manshad, M. Riazi, J.A. Ali, S.M. Sajadi, A. Keshavarz
CuO/TiO₂/PAM as a novel introduced hybrid agent for water-oil interfacial
tension and wettability optimization in chemical enhanced oil recovery
Energy Fuel, 33 (11) (2019), pp. 10547-10560
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [2] M.H. Behdadfar, J.J. Sheng, E. Esmailnezhad
Designing effective enhanced oil recovery fluid: combination of graphene
oxide, D118 SuperPusher, and Chuback surfactant
J. Mol. Liq., 390 (2023), Article 123081
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [3] M. Sepahvand, H. Ghalelavi, F.S. Goharrizi, M. Schaffie, A. Hemmati-Sarapardeh
Integrating low salinity water, surfactant solution, and functionalized
magnetite nanoparticles with natural acidic groups for enhanced oil
recovery: interfacial tension study
J. Mol. Liq., 405 (2024), Article 124944
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [4] E. Jafarbeigi, S. Ayatollahi, Y. Ahmadi, M. Mansouri, F. Dehghani

Identification of novel applications of chemical compounds to change the wettability of reservoir rock: a critical review

J. Mol. Liq., 371 (2022), Article 121059

[Google Scholar ↗](#)

- [5] G. Zargar, T. Arabpour, A. Khaksar-Manshad, J.A. Ali, S.M. Sajadi, A. Keshavarz, A.H. Mohammadi
Experimental investigation of the effect of green TiO₂/quartz nanocomposite on interfacial tension reduction, wettability alteration, and oil recovery improvement
Fuel, 263 (2020), Article 116599



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [6] A. Muggeridge, A. Cockin, K. Webb, H. Frampton, I. Collins, T. Moulds, P. Salino
Recovery rates, enhanced oil recovery and technological limits
Philos. Trans R Soc. A., 372 (2014), p. 20120320

[Crossref ↗](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [7] P. Koreh, M. Lashkarbolooki, M. Peyravi, H.N. Al-Saedi
Applying membrane technology for preparing low salinity for hybrid nanofluid, surfactant, and alkaline enhanced oil recovery process
Fuel, 390 (2025), Article 134713



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [8] A. Rezaei, M. Riazi, M. Escrochi, R. Elhaei
Integrating surfactant, alkali and nano-fluid flooding for enhanced oil recovery: a mechanistic experimental study of novel chemical combinations
J. Mol. Liq., 308 (2020)

[Google Scholar ↗](#)

- [9] M. Almahfood, B. Bai

The synergistic effects of nanoparticle-surfactant nanofluids in EOR applications

J. Pet. Sci. Eng., 171 (2018), pp. 196-210



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[10]

A. Mandal

Chemical flood enhanced oil recovery: a review

Int J. Oil Gas Coal Technol., 9 (3) (2015)

[Google Scholar ↗](#)

[11]

M. Mohajeri, M. Hemmati, A.S. Shekarabi

An experimental study on using a nanosurfactant in an EOR process of heavy oil in a fractured micromodel

J. Pet. Sci. Eng., 126 (2015), pp. 162-173



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[12]

F. Navaie, E. Esmailnezhad, H.J. Choi

Effect of rheological properties of polymer solution on polymer flooding characteristics

Polymers, 14 (24) (2022), p. 5555

[Crossref ↗](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[13]

A. Ghaedi, M. Nabipour, A. Azdarpour, A. Gandomkar, Mechanistic investigation of using low salinity alkaline surfactant solutions in carbonate reservoirs based on polarity of crude oil, J. Mol. Liq. 390 (2023) 123098.

[Google Scholar ↗](#)

[14]

M. Ghorbani, M. Hosseini, M. Lashkarbolooki, G.N. Darzi

Modification and mechanistic study of fluid/fluid and carbonate rock/fluid interfaces using a glycolipid biosurfactant from an indigenous strain of *Gordonia terrae* for MEOR applications

J. Mol. Liq., 414 (2024), Article 126198



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

[15]

H. Garmsiri, S. Jahani, Y. Kazemzadeh, M. Sharifi, M. Riazi, R. Azin

Stability of the emulsion during the injection of anionic and cationic surfactants in the presence of various salts

Sci. Rep., 13 (1) (2023), p. 11337

[View in Scopus](#) [Google Scholar](#)

[16]

M. Shayan-Nasr, E. Esmailnezhad, H.J. Choi

Effect of silicon-based nanoparticles on enhanced oil recovery

J. Taiwan Inst. Chem. Eng., 122 (2021), pp. 241-259



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

[17]

Y. Ahmadi, M. Hemmati, B. Vaferi, A. Gandomkar

Applications of nanoparticles during chemical enhanced oil recovery: a review of mechanisms and technical challenges

J. Mol. Liq., 126287 (2024)

[Google Scholar](#)

[18]

E. Jafarbeigi, A. Mohammadidoust, B. Ranjbar

A review on applications of nanoparticles in the enhanced oil recovery in carbonate reservoirs

J. Pet. Sci. Technol., 15 (40) (2020)

[Google Scholar](#)

[19]

J. Wackerlig, P.A. Lieberzeit

Molecularly imprinted polymer nanoparticles in chemical sensing—synthesis, characterisation and application

Sensors Actuators B Chem., 207 (2015), pp. 144-157



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [20] M. Hasanzadeh, F. Mokhtari, N. Shadjou, A. Eftekhari, A. Mokhtarzadeh, V. Jouyban-Gharamaleki, S. Mahboob
Poly arginine-graphene quantum dots as a biocompatible and non-toxic nanocomposite: layer-by-layer electrochemical preparation, characterization and non-invasive malondialdehyde sensory application in exhaled breath condensate
Mater SciEng C., 75 (2017), pp. 247-258
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [21] X. Lu, D. Wu, Z. Li, G. Chen
Chapter 7—polymer nanoparticles
ProgMolBiolTransl Sci., 104 (2011), pp. 299-323
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [22] H. ShamsiJazeyi, C.A. Miller, M.S. Wong, J.M. Tour, R. Verduzco
Polymer-coated nanoparticles for enhanced oil recovery
J ApplPolym Sci., 131 (15) (2014)
[Google Scholar ↗](#)
- [23] E. Jafarbeigi, M. Shahini-Nia, M. Mansouri, T. Kikhavani, N. Setareshenas
CuO/Al₂O₃/carbomer as a new hybrid agent for wettability alteration and oil recovery in carbonate reservoirs
J. Mol. Liq., 425 (2025), Article 127270
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [24] M. Shafiei, Y. Kazemzadeh, D.A. Martyushev, Z. Dai, M. Riazi
Effect of chemicals on the phase and viscosity behavior of water in oil emulsions
Sci. Rep., 13 (1) (2023), p. 4100
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [25] F. Salimi, E. Jafarbeigi, C. Karami, E. Khodapanah

Synthesis of cost-effective Si-CQD for effective oil separation from core rock

J. Mol. Liq., 394 (2024), Article 123722



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [26] A.O. Gbadamosi, R. Junin, M.A. Manan, A. Agi, J.O. Oseh, J. Usman
Synergistic application of aluminium oxide nanoparticles and oilfield polyacrylamide for enhanced oil recovery
J. Pet. Sci. Eng., 182 (2019)
[Google Scholar ↗](#)
- [27] N. Kumar, S. Ali, A. Kumar, A. Mandal
Design and formulation of surfactant stabilized O/W emulsion for application in enhanced oil recovery: effect of pH, salinity and temperature, oil gas
Sci. Technol., 75 (72) (2020)
[Google Scholar ↗](#)
- [28] N. Pal, N. Kumar, A. Mandal
Stabilization of dispersed oil droplets in nanoemulsions by synergistic effects of the Gemini surfactant, PHPA polymer, and silica nanoparticle
Langmuir, 35 (7) (2019), pp. 2655-2667
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [29] R. Saha, R.V.S. Uppaluri, P. Tiwari
Impact of natural surfactant (Reetha), polymer (xanthan gum), and silica nanoparticles to enhance heavy crude oil recovery
Energy Fuel, 33 (5) (2019), pp. 4225-4236
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [30] A. Sekerbayeva, P. Pourafshary, M.R. Hashmet

Application of anionic surfactant/engineered water hybrid EOR in carbonate formations: an experimental analysis

Petroleum, 8 (4) (2022), pp. 466-475



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [31] D. Joshi, D.N. Ramesh, S. Prakash, R.K. Saw, N.K. Maurya, K.B. Rathi, S. Mitra, O.P. Sinha, P.K. Bikkina, A. Mandal

Formulation and characterization of polymer and nanoparticle-stabilized anionic surfactant foam for application in enhanced oil recovery

Surf Interfaces, 56 (2025), Article 105615



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [32] X. Zhong, C. Li, Y. Li, H. Pu, Y. Zhou, J.X. Zhao

Enhanced oil recovery in high salinity and elevated temperature conditions with a zwitterionic surfactant and silica nanoparticles acting in synergy

Energy Fuel, 34 (3) (2020), pp. 2893-2902

[Crossref ↗](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [33] X. Zhong, C. Li, H. Pu, Y. Zhou, J.X. Zhao

Increased nonionic surfactant efficiency in oil recovery by integrating with hydrophilic silica nanoparticle

Energy Fuel, 33 (9) (2019), pp. 8522-8529

[Crossref ↗](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [34] J.A. Ali, K. Kolo, A. Khaksar-Manshad, A.H. Mohammadi

Recent advances in application of nanotechnology in chemical enhanced oil recovery: effects of nanoparticles on wettability alteration, interfacial tension reduction, and flooding

Egypt. J. Pet., 27 (4) (2018), pp. 1371-1383



[View PDF](#)

[View article](#)

[Crossref ↗](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [35] I. Nowrouzi, A.H. Mohammadi, A. Khaksar-Manshad
Evaluating the feasibility of using a rapeseed oil-derived anionic polymeric surfactant for enhanced oil recovery from carbonate/sandstone composite reservoirs
Can. J. Chem. Eng., 103 (2025), pp. 3071-3091
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [36] M. Fouji, J.J. Sheng, T. Ganat, E. Esmaeilnezhad
A performance evaluation of magnetic nanoparticle, biosurfactant, and polymer fluids for enhanced oil recovery under magnetic field stimulation
J. Mol. Liq., 127823 (2025)
[Google Scholar ↗](#)
- [37] B. Ahmadi, E. Sahraei, A.H. Mohammadi
Investigation of the synergistic effects of SiO₂ and Al₂O₃ nanoparticles with SDS and CTAB surfactants on the stability and improving phase behavior of water-oil emulsions
Colloids Surf. A Physicochem. Eng. Asp., 705 (2) (2024), Article 135726
[Google Scholar ↗](#)
- [38] J.A. Ali, K. Kolo, A. Khaksar-Manshad, K.D. Stephen
Potential application of low-salinity polymeric-nanofluid in carbonate oil reservoirs: IFT reduction, wettability alteration, rheology and emulsification characteristics
J. Mol. Liq., 284 (2019), pp. 735-747
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [39] A. Rezaei, M. Abdi-Khangah, A. Mohebbi, A. Tatar, A.H. Mohammadi
Using surface modified clay nanoparticles to improve rheological behavior of hydrolyzed polyacrylamide (HPAM) solution for enhanced oil recovery with polymer flooding

J. Mol. Liq., 222 (2016), pp. 1148-1156



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [40] T. Sharma, S. Iglauer, J.S. Sangwai
Silica nanofluids in an oilfield polymer polyacrylamide: interfacial properties, wettability alteration, and applications for chemical enhanced oil recovery

IndEngChem Res., 55 (48) (2016), pp. 12387-12397

[Crossref](#) [View in Scopus](#) [Google Scholar](#)

- [41] S.K. Rai, A. Bera, A. Mandal
Modeling of surfactant and surfactant-polymer flooding for enhanced oil recovery using STARS (CMG) software

J. Pet. Explor. Prod. Technol., 5 (2015), pp. 1-11

[Crossref](#) [View in Scopus](#) [Google Scholar](#)

- [42] M.T. Hayavi, Y. Kazemzadeh, M. Riazi
Application of surfactant-based enhanced oil recovery in carbonate reservoirs: a critical review of the opportunities and challenges

Chem. Phys. Lett., 806 (2022), Article 139975

[Google Scholar](#)

- [43] H. Zhang, T.S. Ramakrishnan, A. Nikolov, D. Wasan
Enhanced oil recovery driven by nanofilm structural disjoining pressure: flooding experiments and micro visualization

Energy Fuel, 30 (4) (2016), pp. 2771-2779

[Crossref](#) [View in Scopus](#) [Google Scholar](#)

- [44] Y. Kazemzadeh, S. Shojaei, M. Riazi, M. Sharifi
Review on application of nanoparticles for EOR purposes: a critical review of the opportunities and challenges

Chin. J. Chem. Eng., 27 (2) (2019), pp. 237-246

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [45] J. Yu, C. An, D. Mo, D. Liu, R. Lee
Foam Mobility Control for Nanoparticle-Stabilized CO₂ Foam, SPE Paper
153336

Improved Oil Recovery Symposium, Tulsa, Oklahoma, USA (2012)

[Google Scholar ↗](#)

- [46] E. Jafarbeigi, Y. Ahmadi, M. Mansouri, S. Ayatollahi
Experimental core flooding investigation of new ZnO- γ -Al₂O₃
nanocomposites for enhanced oil recovery in carbonate reservoirs

ACS Omega, 7 (43) (2022), pp. 39107-39121

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [47] Y. Ahmadi, M. Mansouri, E. Jafarbeigi
Improving simultaneous water alternative associate gas tests in the
presence of newly synthesized γ -Al₂O₃/ZnO/urea nano-composites: an
experimental core flooding tests

ACS Omega, 8 (1) (2022), pp. 1443-1452

[Google Scholar ↗](#)

- [48] A. Omid, A. Khaksar-Manshad, S. Moradi, J.A. Ali, S.M. Sajadi, A. Keshavarz
Smart- and nano-hybrid chemical EOR flooding using Fe₃O₄/eggshell
nanocomposites

J. Mol. Liq., 316 (2020), Article 113880

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [49] M.J. Nazarahari, A. Khaksar-Manshad, M. Ali, J.A. Ali, A. Shafiei, S.M. Sajadi, S. Moradi, S.
Iglauer, A. Keshavarz
Impact of a novel biosynthesized nanocomposite
(SiO₂@Montmorilant@xanthan) on wettability shift and interfacial
tension: applications for enhanced oil recovery

Fuel, 298 (2021), Article 120773



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [50] J.A. Ali, K. Kolo, A. Khaksar-Manshad, K.D. Stephen
Emerging applications of TiO₂/SiO₂/poly(acrylamide) nanocomposites
within the engineered water EOR in carbonate reservoirs
J. Mol. Liq., 322 (2021), Article 114943



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [51] M.J. Nazarahari, A. Khaksar-Manshad, S. Moradi, A. Shafiei, J.A. Ali, S.M. Sajadi, A. Keshavarz
Synthesis, characterization, and assessment of a CeO₂@Nanoclay
nanocomposite for enhanced oil recovery
Nanomaterials, 10 (11) (2020), p. 2280

[Google Scholar ↗](#)

- [52] ASTM D2007–91: Standard Test Method for Characteristic Groups in Rubber Extender and
Processing Oils and Other Petroleum-Derived Oils by the Clay-Gel Absorption
Chromatographic Method; Book of Standards 5.01, American Society for Testing and
Materials, Philadelphia, PA (1993)

[Google Scholar ↗](#)

- [53] U. Zulfiqar, T. Subhani, H.S. Wilayat
Synthesis of silica nanoparticles from sodium silicate under alkaline
conditions
J. Sol-Gel Sci. Technol., 77 (3) (2016), pp. 753-758

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [54] R. Chaturvedi, T. Sharma
Rheological analysis and EOR potential of surfactant treated single-step
silica nanofluid at high temperature and salinity
J. Pet. Sci. Eng., 196 (2021), Article 107704

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [55] A. Giampiccolo, R. Ball, M. Ansell, D. Maskell
Synthesis of co-Doped TiO₂ Nanostructures for Novel Photocatalytic Coatings
ENERGY FORUM on Advanced Building Skins, Italy (2014)
[Google Scholar ↗](#)
- [56] K. Cacia, F. Ordoñez, C. Zapata, B. Herrera, E. Pabón, R. Buitrago-Sierra
Surfactant concentration and pH effects on the zeta potential values of alumina nanofluids to inspect stability
Colloids Surf. A Physicochem. Eng. Asp., 583 (2019), Article 123960
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [57] M.J. Ismay, E. Doroodchi, B. Moghtaderi
Effects of colloidal properties on sensible heat transfer in water-based titaniananofluids
ChemEng Res Des., 91 (2013), pp. 426-436
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [58] E. Jafarbeigi, E. Kamari, F. Salimi, A. Mohammadidoust
Experimental study of the effects of a novel nanoparticle on enhanced oil recovery in carbonate porous media
J. Pet. Sci. Eng., 195 (2020), Article 107602
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [59] B. YadaliJamaloei, R. Kharrat
Analysis of microscopic displacement mechanisms of dilute surfactant flooding in oil-wet and water-wet porous media
Transp. Porous Media, 81 (2009), p. 1
[Google Scholar ↗](#)

- [60] C. Song, H. Yoo, J. Lee
Effects of nano-smart water on enhanced oil recovery in carbonate reservoirs: interfacial tension reduction and wettability alteration
Colloids Surf. A Physicochem. Eng. Asp., 696 (2024), Article 134362
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [61] W. Anderson
Wettability literature survey-part 2: wettability measurement
J. Pet. Sci. Technol., 38 (1986), pp. 241-246
[Google Scholar ↗](#)
- [62] L. Wang, G. Xie, X. Mi, B. Zhang, Y. Du, Q. Zhu, Z. Yu
Surface-modified TiO₂@SiO₂ nanocomposites for enhanced dispersibility and optical performance to apply in the printing process as a pigment
ACS Omega., 8 (22) (2023), pp. 20116-20124
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [63] M. Bellardita, M. Addamo, A. Di-Paola, G. Marci, L. Palmisano, L. Cassar, M. Borsa
Photocatalytic activity of TiO₂/SiO₂ systems
J. Hazard. Mater., 174 (2010), pp. 707-713
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [64] R. Fattahi, M. Lashkarbolooki, R. Abedini, H. Younesi
Investigating the synergistic/antagonistic effects of mixing SiO₂ nanoparticles and ionic liquid, nonionic emulsifier, and Gemini surfactants on the main mechanisms of crude oil production
Energy Fuel, 37 (19) (2023), pp. 14741-14751
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [65] M. Ramezani, R. Abedini, M. Lashkarbolooki
Experimental study about the effect of SiO₂ nanoparticle in surfactant performance on IFT reduction and wettability alteration

ChemEng Res Des., 192 (2023), pp. 350-361



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [66] R. Fattahi, M. Lashkarbolooki, R. Abedini, H. Younesi
Analysis of the interfacial tension of cationic imidazolium-based ionic liquid, twin-branched tailed anionic surfactant, and a non-ionic emulsifier in the presence of SiO₂ nanoparticle and amphiphilic oleic components using response surface method

J. Mol. Liq., 381 (2023), Article 121799



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [67] S. Habibi, A. Jafari, Z. Fakhroueian
Wettability alteration analysis of smart water/novel functionalized nanocomposites for enhanced oil recovery

Pet. Sci., 17 (2020), pp. 1318-1328

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [68] M. Shirazi, S. Kord, Y. Tamsilian
Novel smart water-based titaniananofluid for enhanced oil recovery

J. Mol. Liq., 296 (2019), Article 112064



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [69] N. LezorgiaNekabari, A. Barifcani, M. Sarmadivaleh, S. Iglauer
Nanofluids as Novel Alternative Smart Fluids for Reservoir Wettability Alteration

Novel Nanomaterials-Synthesis and Applications, IntechOpen (2017)

[Google Scholar ↗](#)

[View Abstract](#)

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